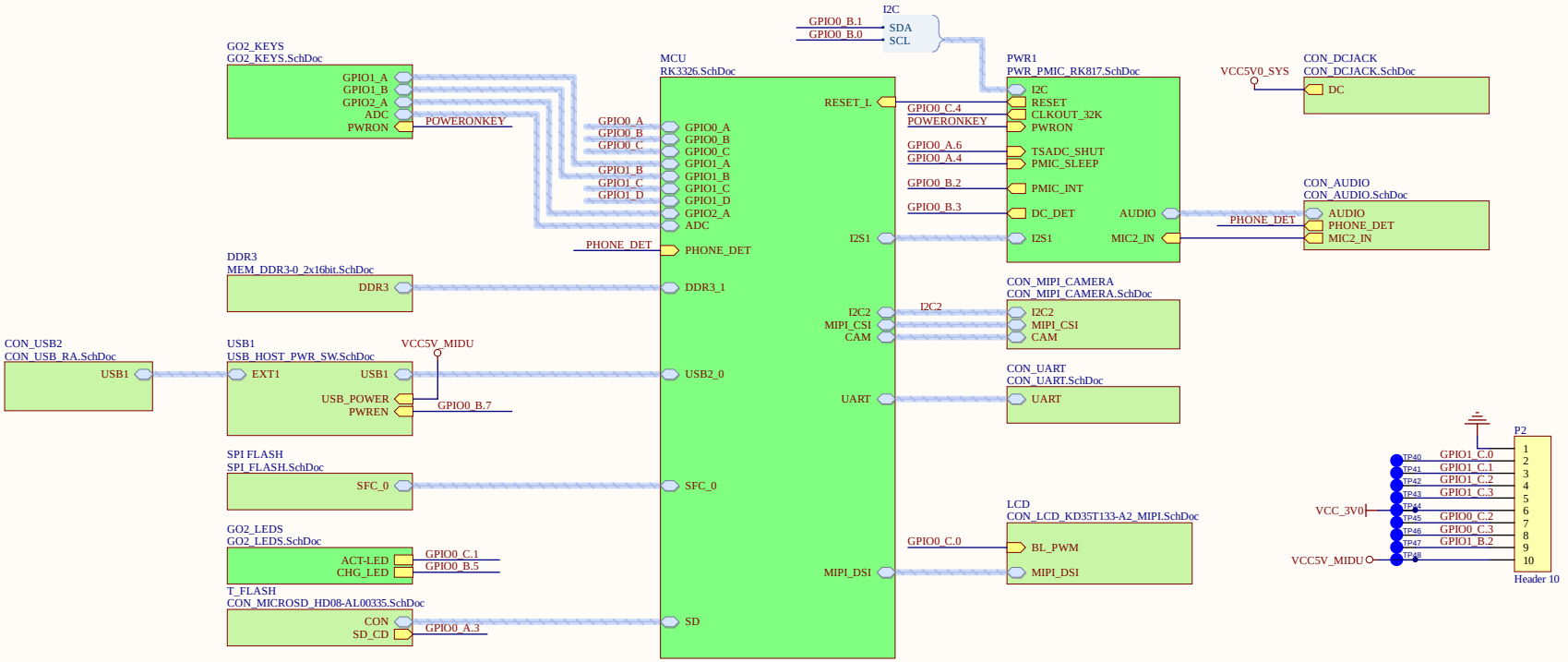
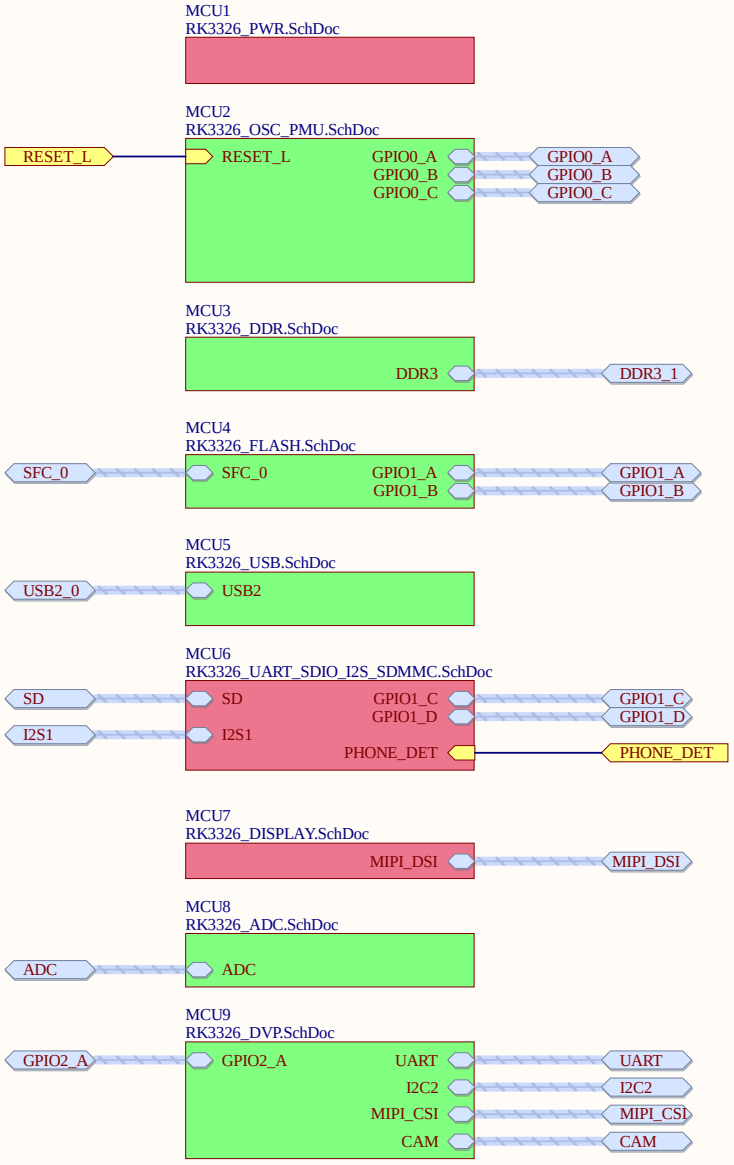


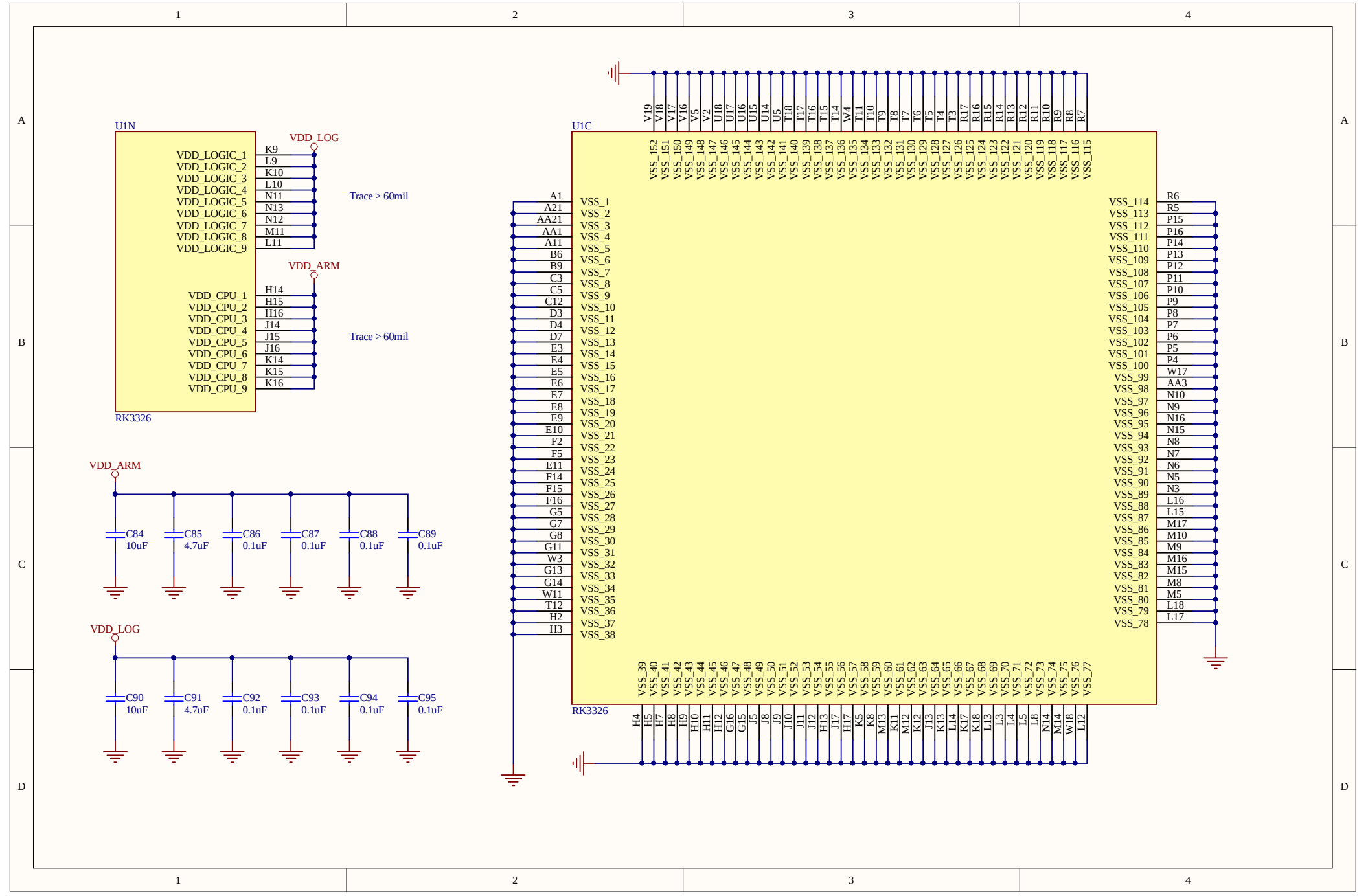

| | | | |
|---|---|---|---|
| Title | GO-ADVANCE-MAIN | | |
| Size: A3 | Number: 2. | Revision: 1.0 | Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea |
| Date: 12/6/2019 | Time: 4:09:44 PM | Sheet 2 of 24 | |
| File: GO2-MAIN.SchDoc | | Designed by: ruppi.kim@hardkernel.com | |

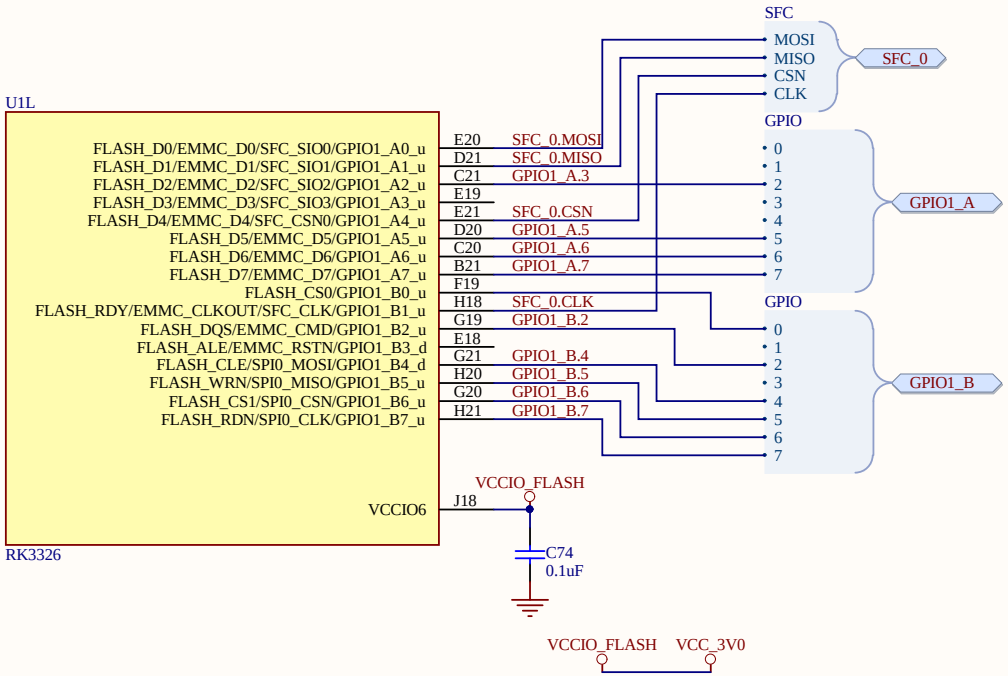
ODROID



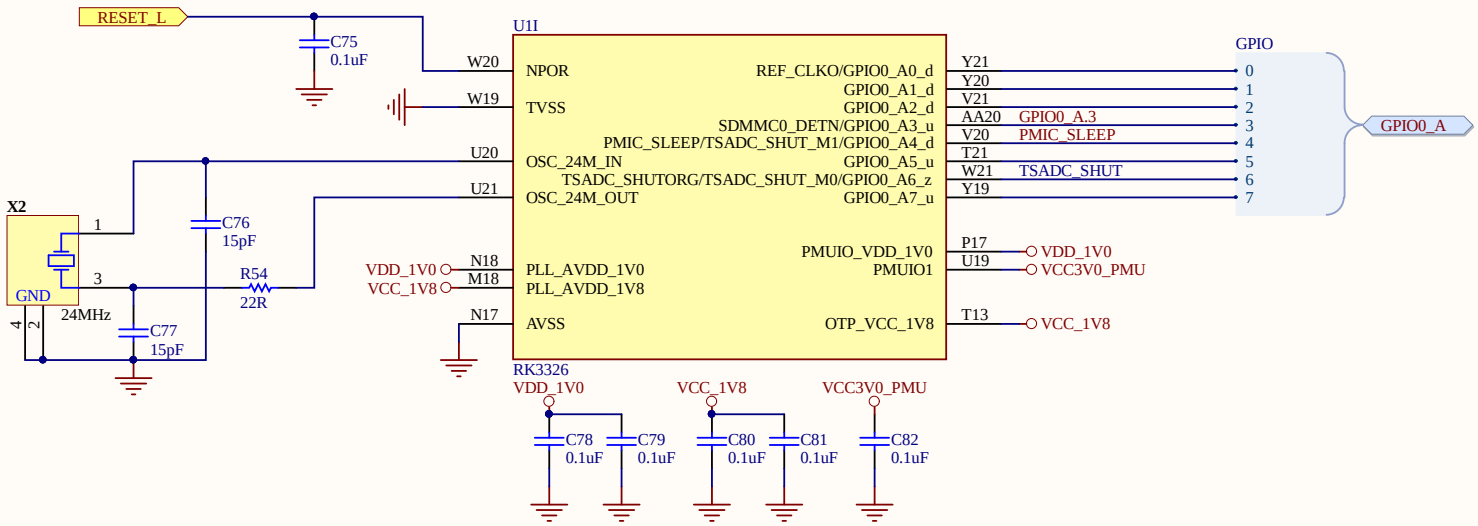

| | | | |
|---|---|---|---|
| Title | ***RK3326*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 3. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:44 PM | Sheet 3 of 24 | | |
| File: RK3326.SchDoc | | Designed by: *@hardkernel.com | |

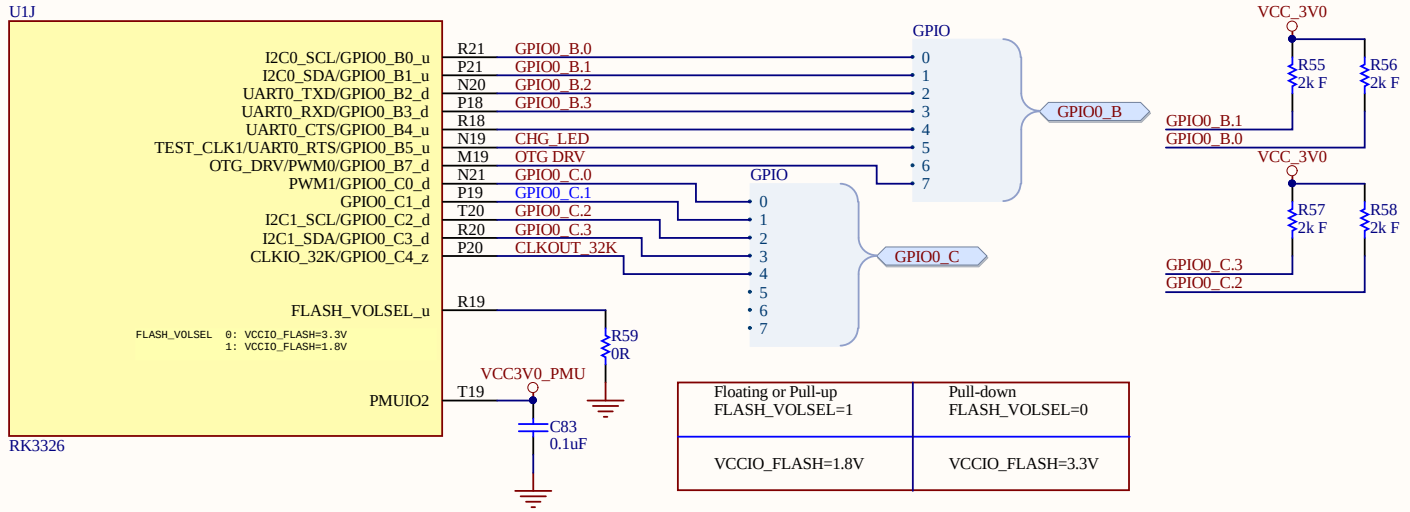
ODROID


U1N
VDD_LOGIC_1
VDD_LOGIC_2
VDD_LOGIC_3
VDD_LOGIC_4
VDD_LOGIC_5
VDD_LOGIC_6
VDD_LOGIC_7
VDD_LOGIC_8
VDD_LOGIC_9
VDD_LOG
K9
L9
K10
L10
N11
N13
N12
M11
L11
Trace > 60mil
VDD_CPU_1
VDD_CPU_2
VDD_CPU_3
VDD_CPU_4
VDD_CPU_5
VDD_CPU_6
VDD_CPU_7
VDD_CPU_8
VDD_CPU_9
VDD_ARM
H14
H15
H16
J14
J15
J16
K14
K15
K16
Trace > 60mil
RK3326
VDD_ARM
C84 10uF
C85 4.7uF
C86 0.1uF
C87 0.1uF
C88 0.1uF
C89 0.1uF
VDD_LOG
C90 10uF
C91 4.7uF
C92 0.1uF
C93 0.1uF
C94 0.1uF
C95 0.1uF
U1C
V19 V18 V17 V16 V5 V2 U18 U17 U16 U15 U14 U5 T18 T17 T16 T15 T14 W4 T11 T10 T9 T8 T7 T6 T5 T4 T3 R17 R16 R15 R14 R13 R12 R11 R10 R9 R8 R7
VSS_152 VSS_151 VSS_150 VSS_149 VSS_148 VSS_147 VSS_146 VSS_145 VSS_144 VSS_143 VSS_142 VSS_141 VSS_140 VSS_139 VSS_138 VSS_137 VSS_136 VSS_135 VSS_134 VSS_133 VSS_132 VSS_131 VSS_130 VSS_129 VSS_128 VSS_127 VSS_126 VSS_125 VSS_124 VSS_123 VSS_122 VSS_121 VSS_120 VSS_119 VSS_118 VSS_117 VSS_116 VSS_115
A1 A21 AA21 AA1 A11 B6 B9 C3 C5 C12 D3 D4 D7 E3 E4 E5 E6 E7 E8 E9 E10 F2 F5 E11 F14 F15 F16 G5 G7 G8 G11 W3 G13 G14 W11 T12 H2 H3
VSS_1 VSS_2 VSS_3 VSS_4 VSS_5 VSS_6 VSS_7 VSS_8 VSS_9 VSS_10 VSS_11 VSS_12 VSS_13 VSS_14 VSS_15 VSS_16 VSS_17 VSS_18 VSS_19 VSS_20 VSS_21 VSS_22 VSS_23 VSS_24 VSS_25 VSS_26 VSS_27 VSS_28 VSS_29 VSS_30 VSS_31 VSS_32 VSS_33 VSS_34 VSS_35 VSS_36 VSS_37 VSS_38
VSS_114 VSS_113 VSS_112 VSS_111 VSS_110 VSS_109 VSS_108 VSS_107 VSS_106 VSS_105 VSS_104 VSS_103 VSS_102 VSS_101 VSS_100 VSS_99 VSS_98 VSS_97 VSS_96 VSS_95 VSS_94 VSS_93 VSS_92 VSS_91 VSS_90 VSS_89 VSS_88 VSS_87 VSS_86 VSS_85 VSS_84 VSS_83 VSS_82 VSS_81 VSS_80 VSS_79 VSS_78
R6 R5 P15 P16 P14 P13 P12 P11 P10 P9 P8 P7 P6 P5 P4 W17 AA3 N10 N9 N16 N15 N8 N7 N6 N5 N3 L16 L15 M17 M10 M9 M16 M15 M8 M5 L18 L17
VSS_39 VSS_40 VSS_41 VSS_42 VSS_43 VSS_44 VSS_45 VSS_46 VSS_47 VSS_48 VSS_49 VSS_50 VSS_51 VSS_52 VSS_53 VSS_54 VSS_55 VSS_56 VSS_57 VSS_58 VSS_59 VSS_60 VSS_61 VSS_62 VSS_63 VSS_64 VSS_65 VSS_66 VSS_67 VSS_68 VSS_69 VSS_70 VSS_71 VSS_72 VSS_73 VSS_74 VSS_75 VSS_76 VSS_77
RK3326
H4 H5 H7 H8 H9 H10 H11 H12 G16 G15 J5 J8 J9 J10 J11 J12 H13 J17 H17 K5 K8 M13 K11 M12 K12 J13 K13 L14 K17 K18 L13 L3 L4 L5 L8 N14 M14 W18 L12




| | | | |
|---|---|---|---|
| Title | ***RK3326 FLASH Controller*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 5. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:45 PM | Sheet 5 of 24 | | |
| File: RK3326_FLASH.SchDoc | | Designed by: *@hardkernel.com | |

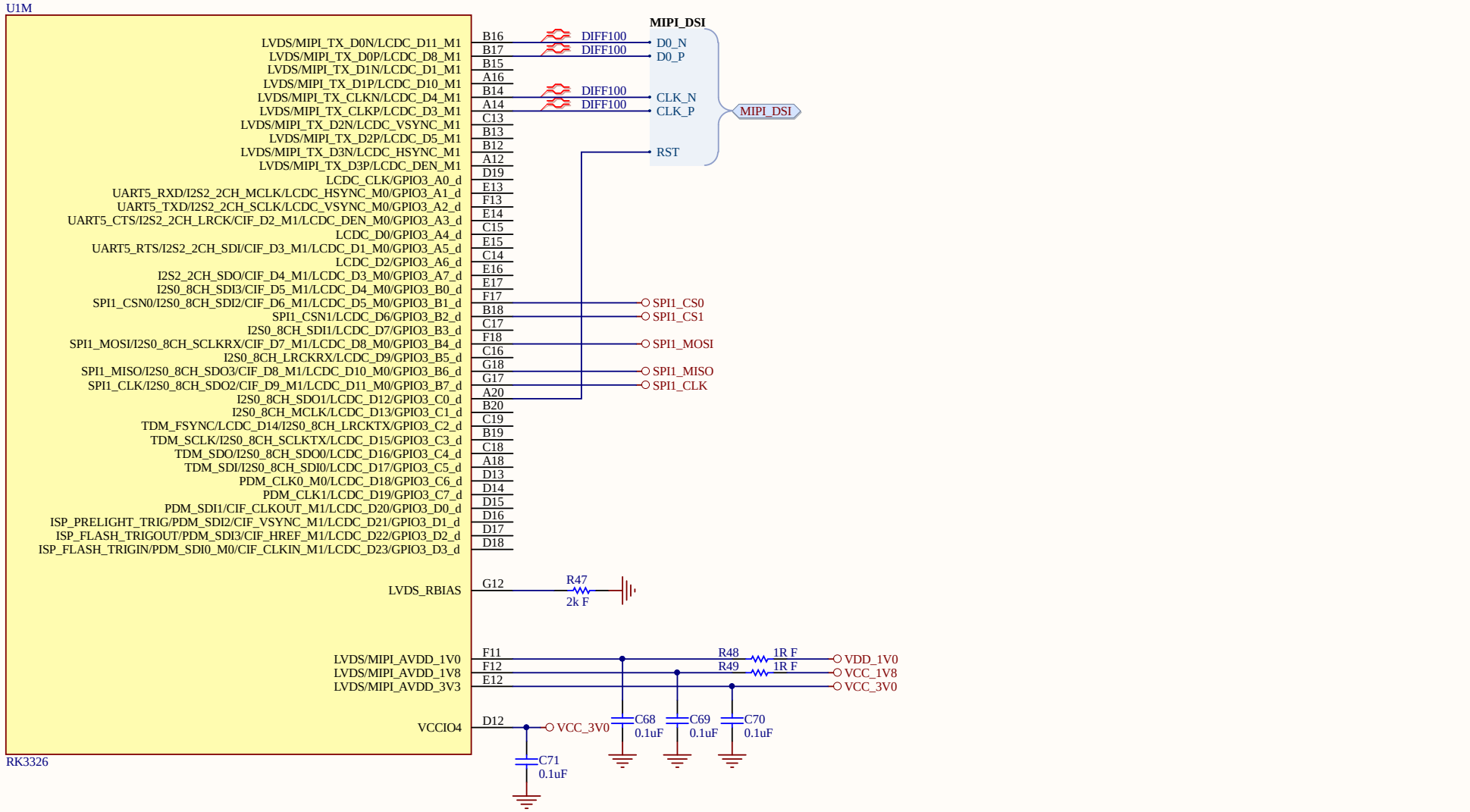



| | | | |
|---|---|---|---|
| Title | RK3326 OSC/PMUIO1/PMUIO2 | | |
| Size: A4 | Number: 6. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:45 PM | Sheet 6 of 24 | |
| File: RK3326_OSC_PMU.SchDoc | | Designed by: *@hardkernel.com | |

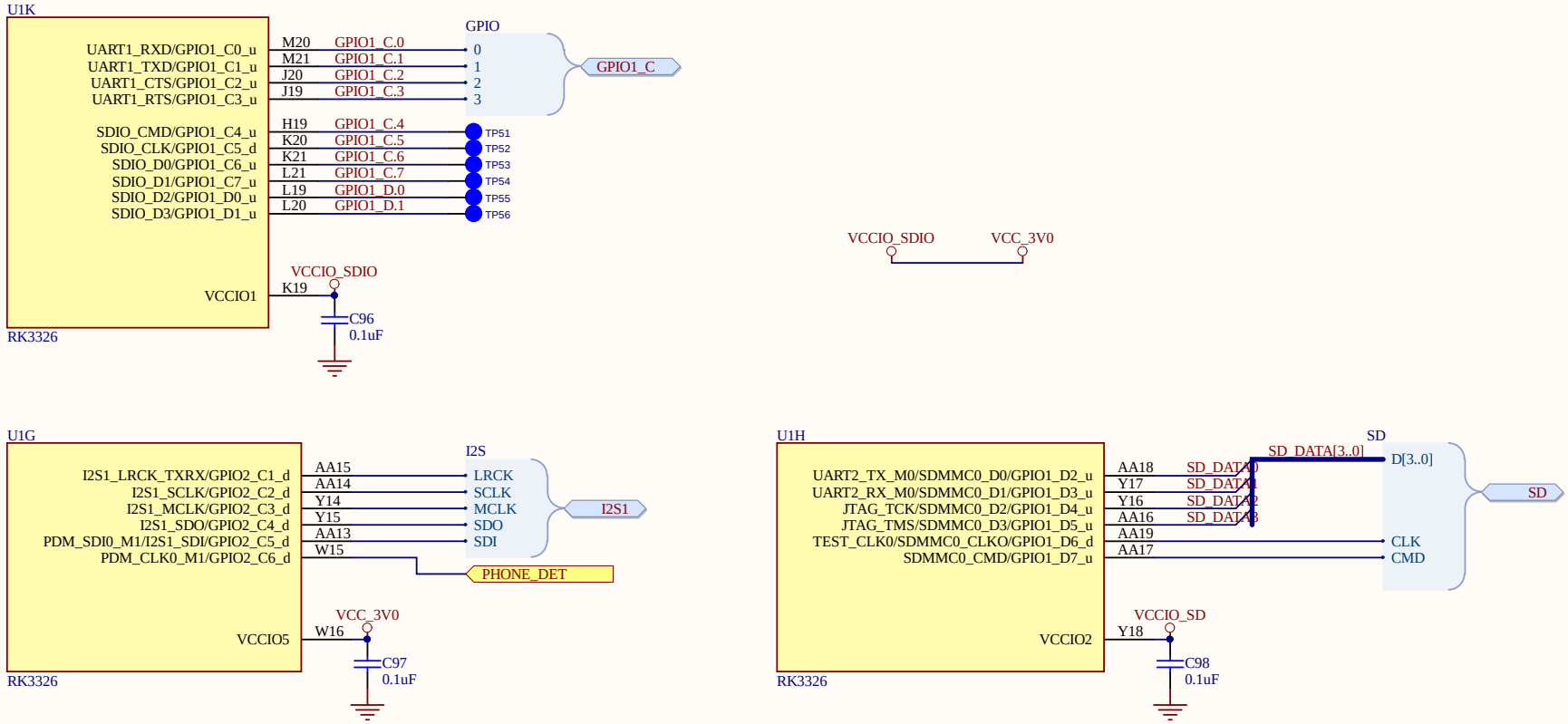
Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea

ODROID



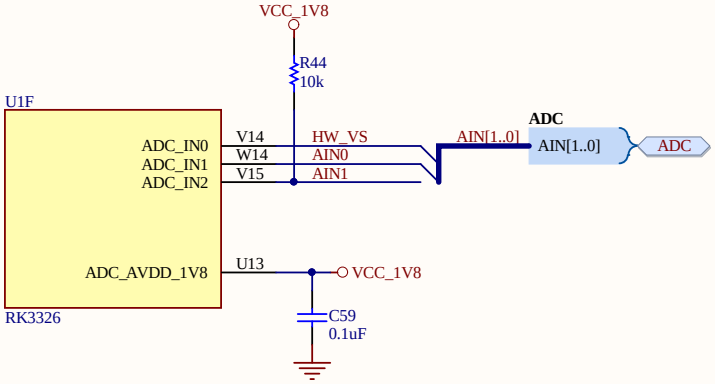

| | | | |
|---|---|---|---|
| Title | ***RK3326 Display Interface*** | | *Hardkernel Co.,LTD.* |
| Size: A4 | Number: 7. | Revision: 1.0 | *475-1 Mananro* |
| Date: 12/6/2019 | Time: 4:09:45 PM | Sheet 7 of 24 | *Manangu, Anyang City* |
| File: RK3326_DISPLAY.SchDoc | | Designed by: *@hardkernel.com | *Gyeonggido* |
| | | | *South Korea* |

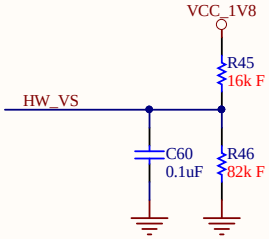
ODROID




| | | | |
|---|---|---|---|
| Title | ***RK3326 UART/SDIO/I2S/SDMMC*** | | *Hardkernel Co.,LTD.* |
| Size: A4 | Number: 8. | Revision: 1.0 | *475-1 Mananro* |
| Date: 12/6/2019 Time: 4:09:45 PM | Sheet 8 of 24 | | *Manangu, Anyang City* |
| File: RK3326_UART_SDIO_I2S_SDMMC.SchDoc | | Designed by: *@hardkernel.com | *Gyeonggido* |
| | | | *South Korea* |

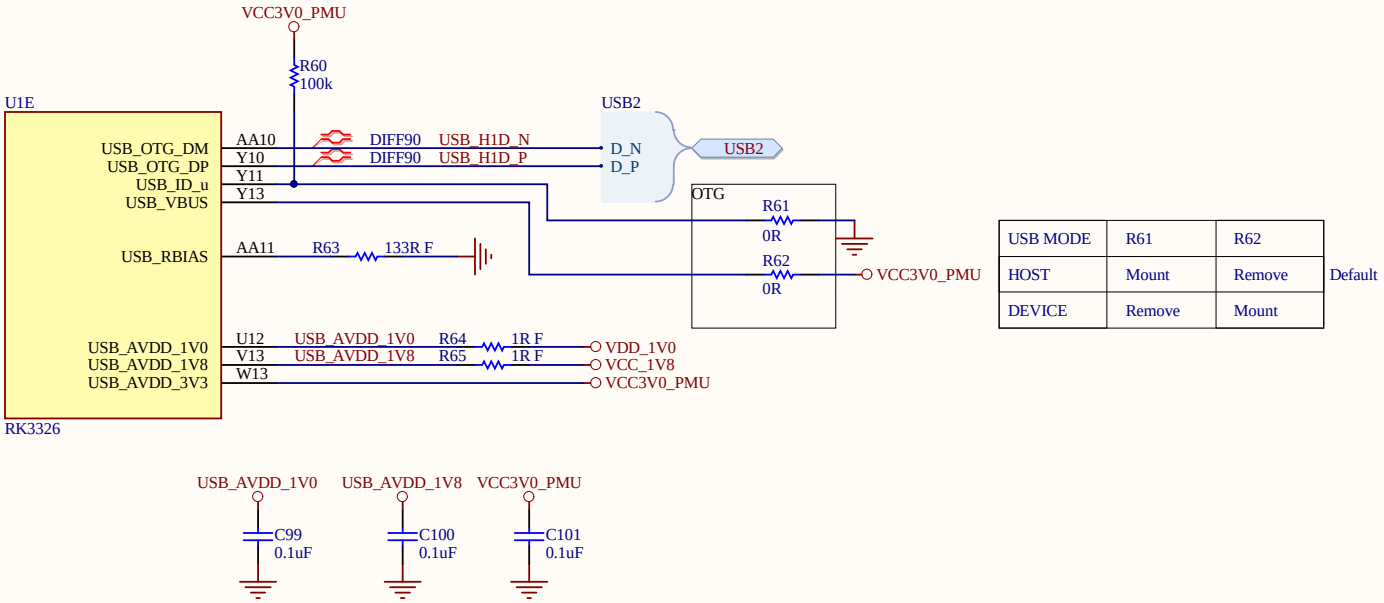
ODROID




| HW ID | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ADC Data | 0 | 95 | 167 | 248 | 334 | 420 | 512 | 603 | 689 | 775 | 856 | 938 | 1023 |
| Pull-up | X | 91K | 82K | 75K | 68K | 56K | 47K | 39K | 33K | 24K | 16K | 8.2K | 100K |
| Pull-down | 100K | 8.2K | 16K | 24K | 33K | 39K | 47K | 56K | 68K | 75K | 82K | 91K | X |
| Version | Reserved | V1 | | | | | | | | | | | |

| | | | |
|---|---|---|---|
| Title | ***RK3326 SARADC*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 9. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:45 PM | Sheet 9 of 24 | | |
| File: RK3326_ADC.SchDoc | | Designed by: *@hardkernel.com | |

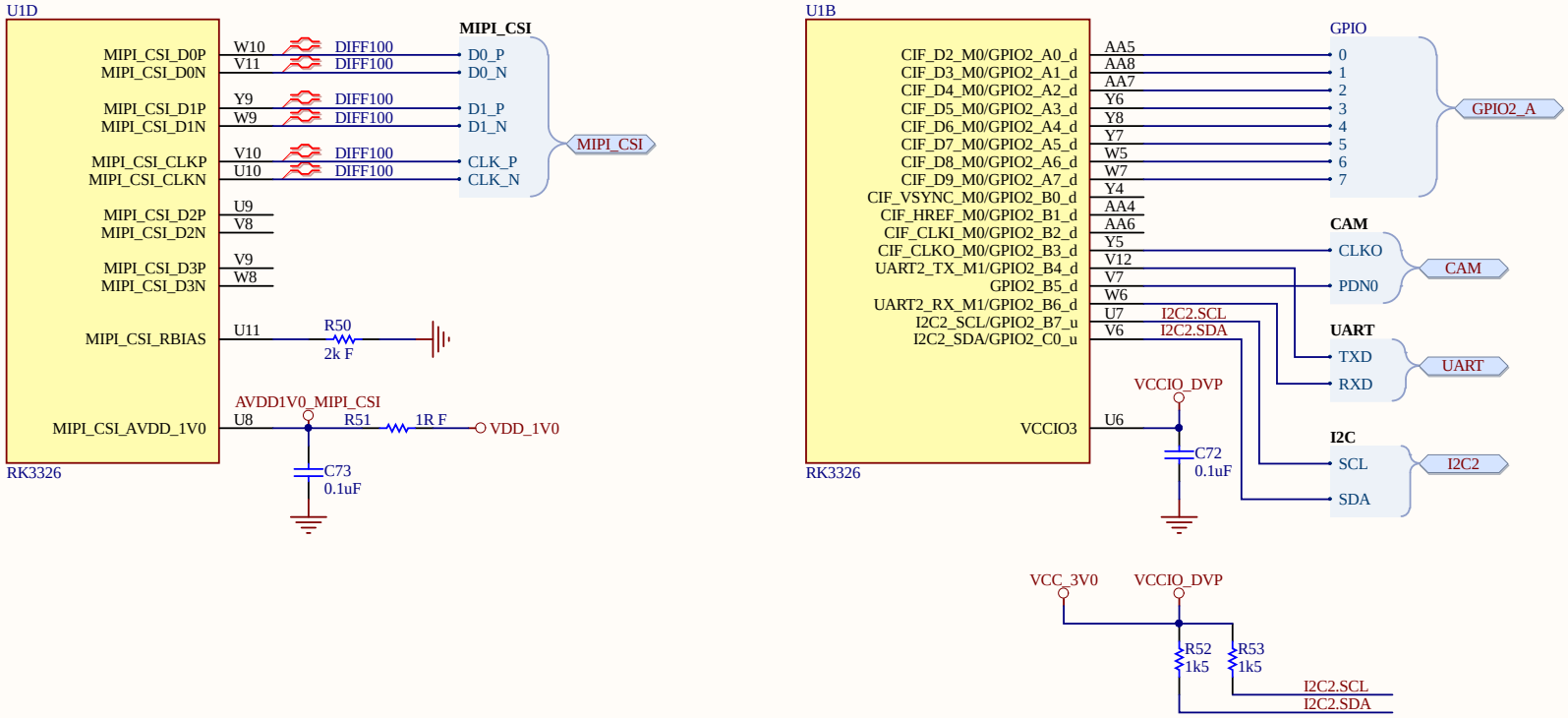



| USB MODE | R61 | R62 | |
|---|---|---|---|
| HOST | Mount | Remove | Default |
| DEVICE | Remove | Mount | |




| | | | |
|---|---|---|---|
| Title | ***RK3326 USB Controller*** | | *Hardkernel Co.,LTD.* |
| Size: A4 | Number: 10. | Revision: 1.0 | *475-1 Mananro* |
| Date: 12/6/2019 | Time: 4:09:45 PM | Sheet 10 of 24 | *Manangu, Anyang City* |
| File: RK3326_USB.SchDoc | | Designed by: *@hardkernel.com | *Gyeonggido* |
| | | | *South Korea* |

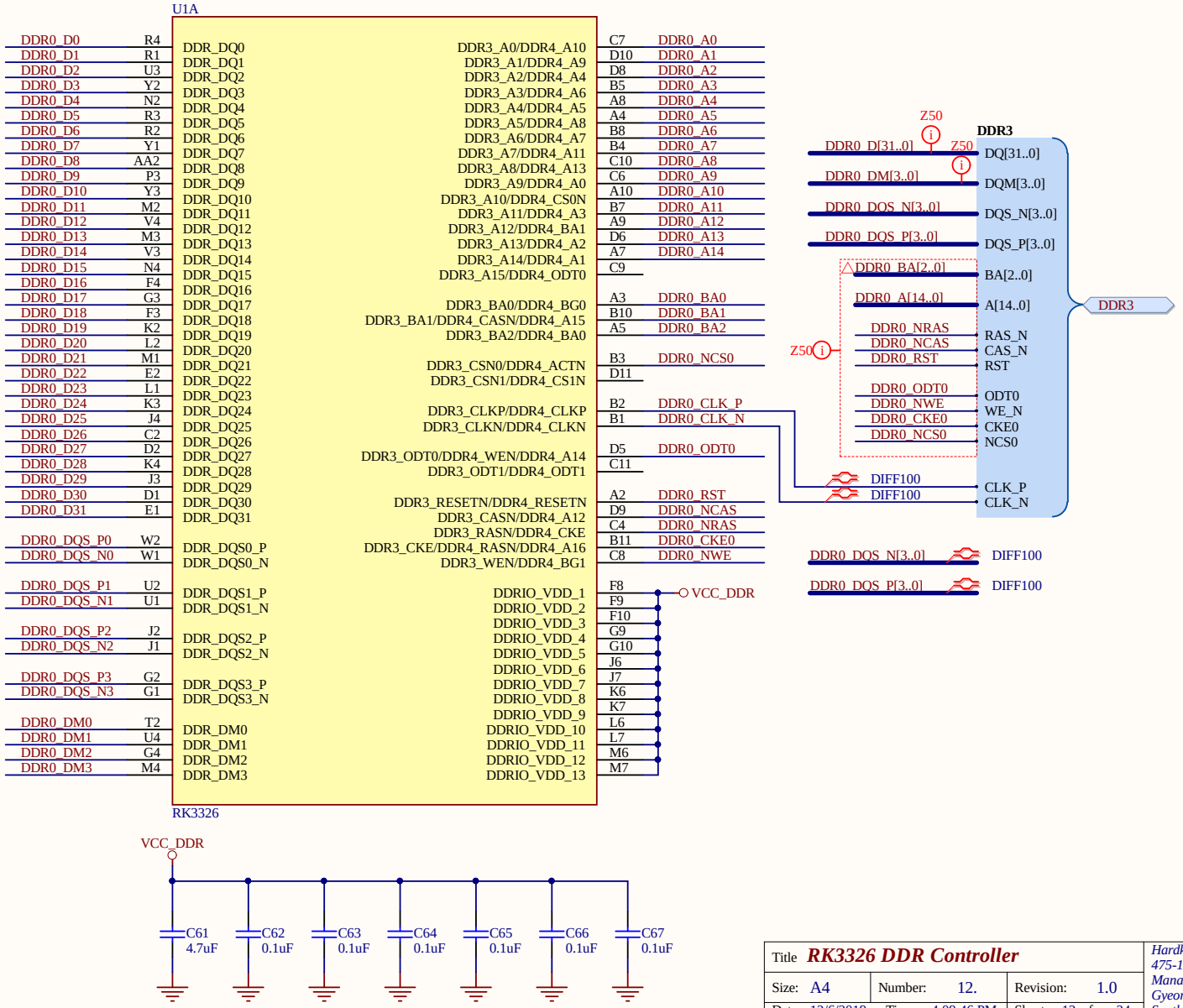
ODROID



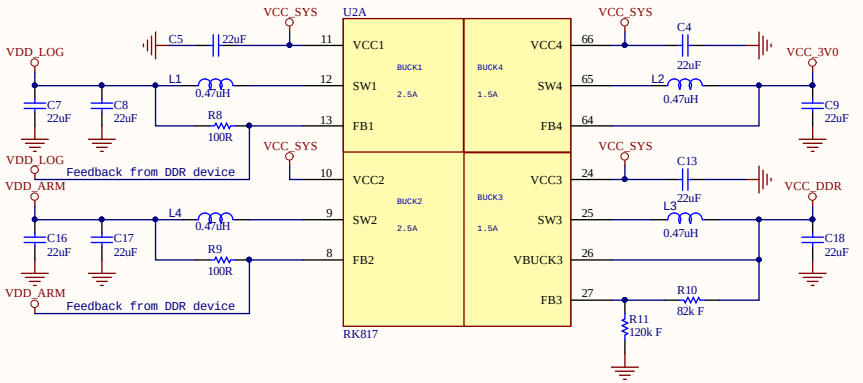

| | | | |
|---|---|---|---|
| Title | ***RK3326 DVP Interface*** | | |
| Size: A4 | Number: 11. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:45 PM | Sheet 11 of 24 | |
| File: RK3326_DVP.SchDoc | | Designed by: *@hardkernel.com | |

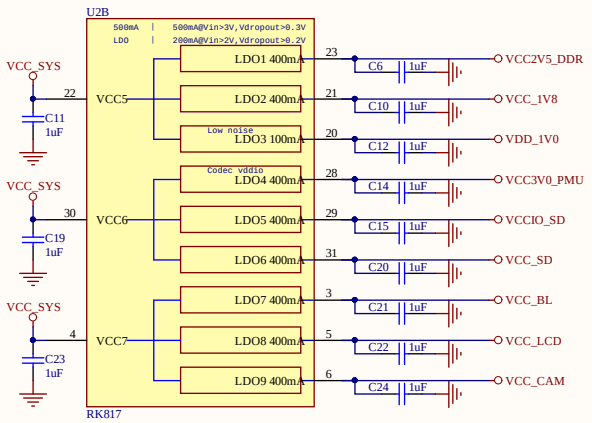
*Hardkernel Co.,LTD.*
*475-1 Mananro*
*Manangu, Anyang City*
*Gyeonggido*
*South Korea*

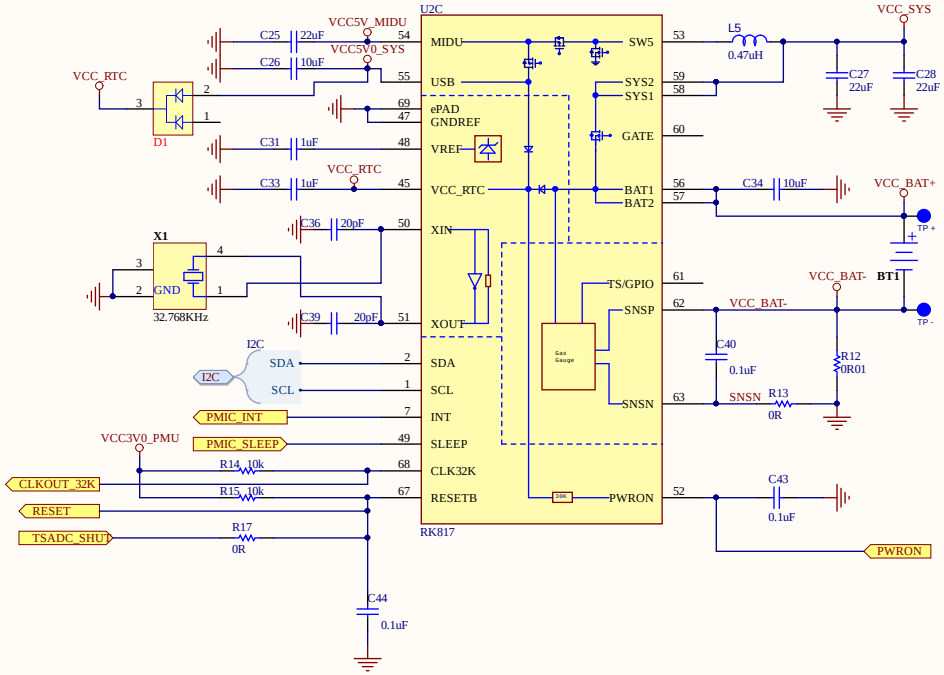
ODROID

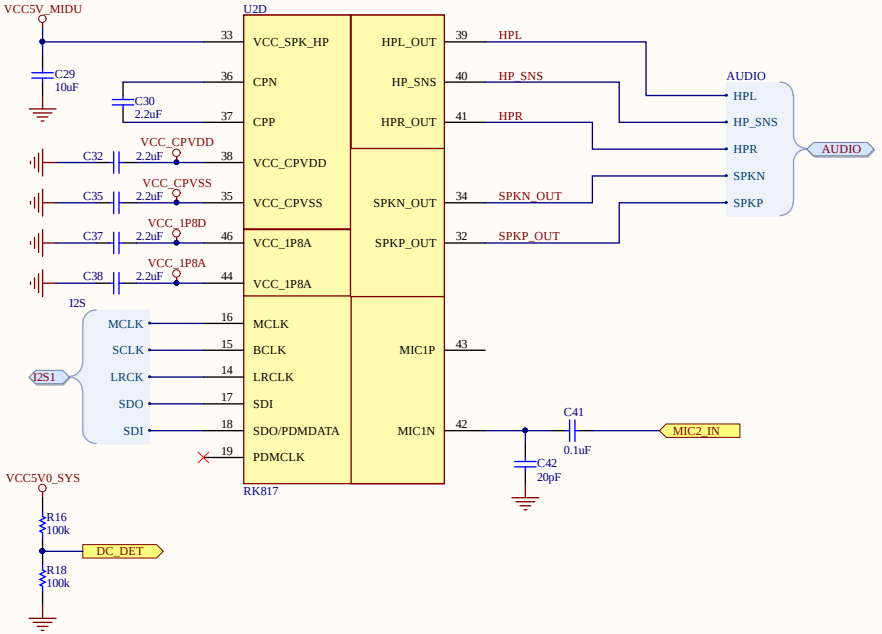



| | | | |
|---|---|---|---|
| Title | **RK3326 DDR Controller** | | Hardkernel Co.,LTD.<br>475-1 Mananro<br>Manangu, Anyang City<br>Gyeonggido<br>South Korea |
| Size: A4 | Number: 12. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:46 PM | Sheet 12 of 24 | | |
| File: RK3326_DDR.SchDoc | | Designed by: *@hardkernel.com | |

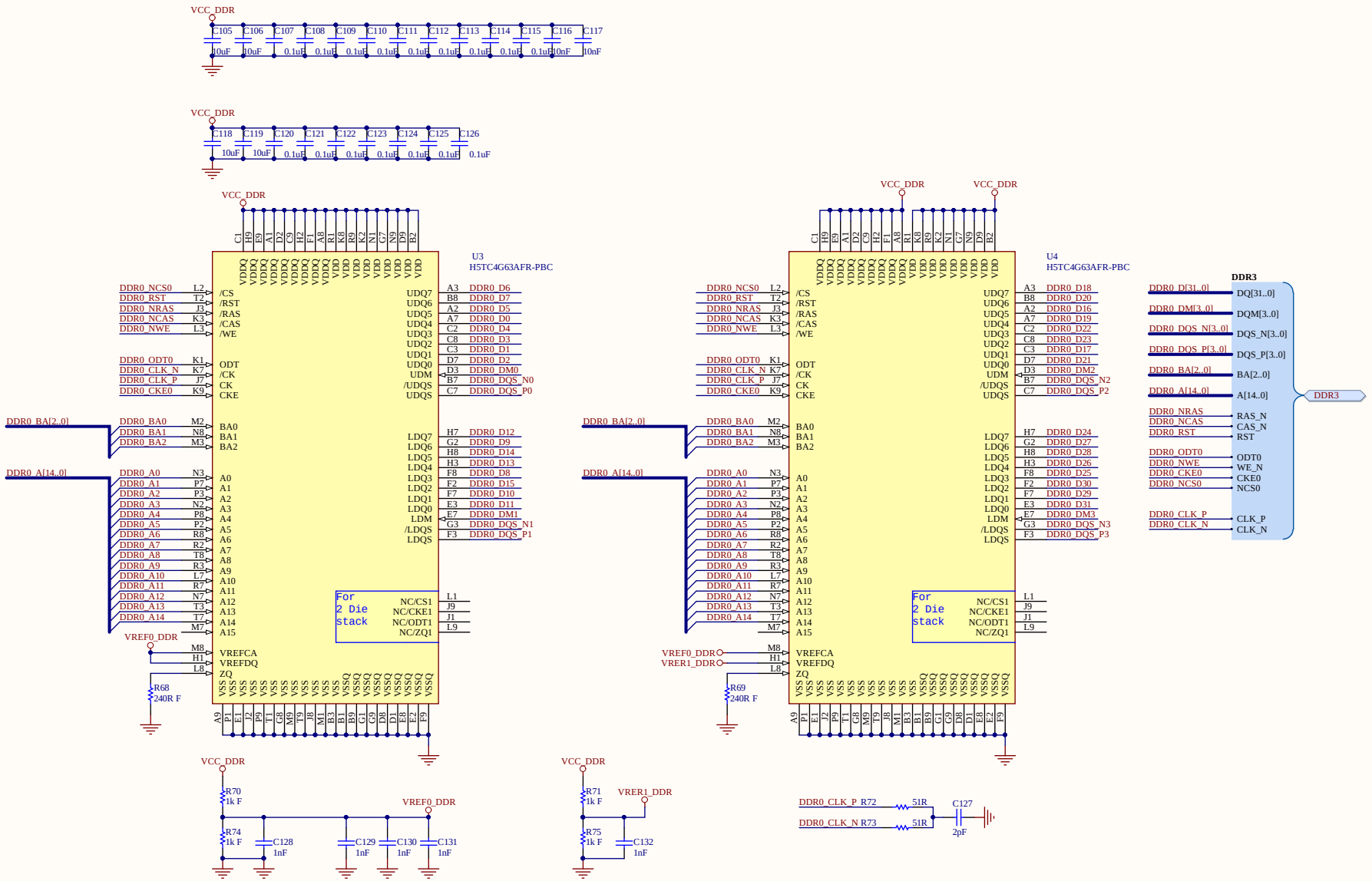


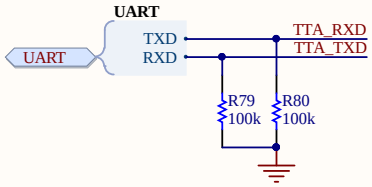



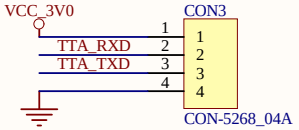




| | | | |
|---|---|---|---|
| Title: RK817 | | | Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea |
| Size: A3 | Number: 13. | Revision: 1.0 | ODROID |
| Date: 12/6/2019 | Time: 4:09:46 PM | Sheet 13 of 24 | |
| File: PWR_PMIC_RK817.SchDoc | | Designed by: ruppi kim@hardkernel.com | |

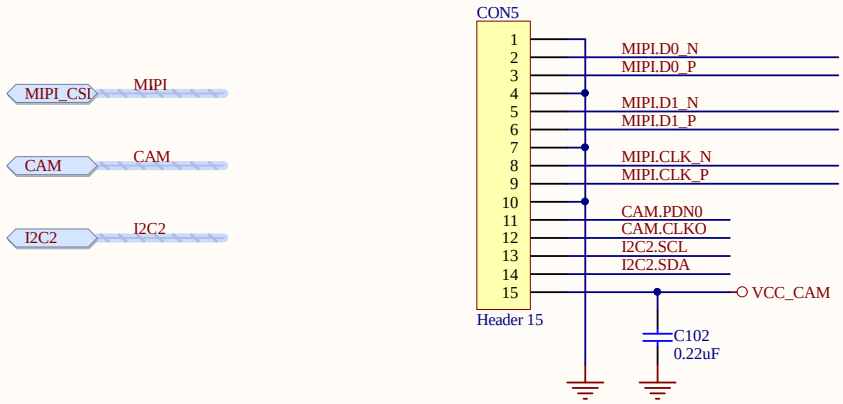



| | | | |
|---|---|---|---|
| Title | *MEM DDR3 2x16bit* | | Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea |
| Size: A3 | Number: 14. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:46 PM | Sheet 14 of 24 | |
| File: MEM_DDR3-0_2x16bit.SchDoc | | Designed by: *@hardkernel.com | |

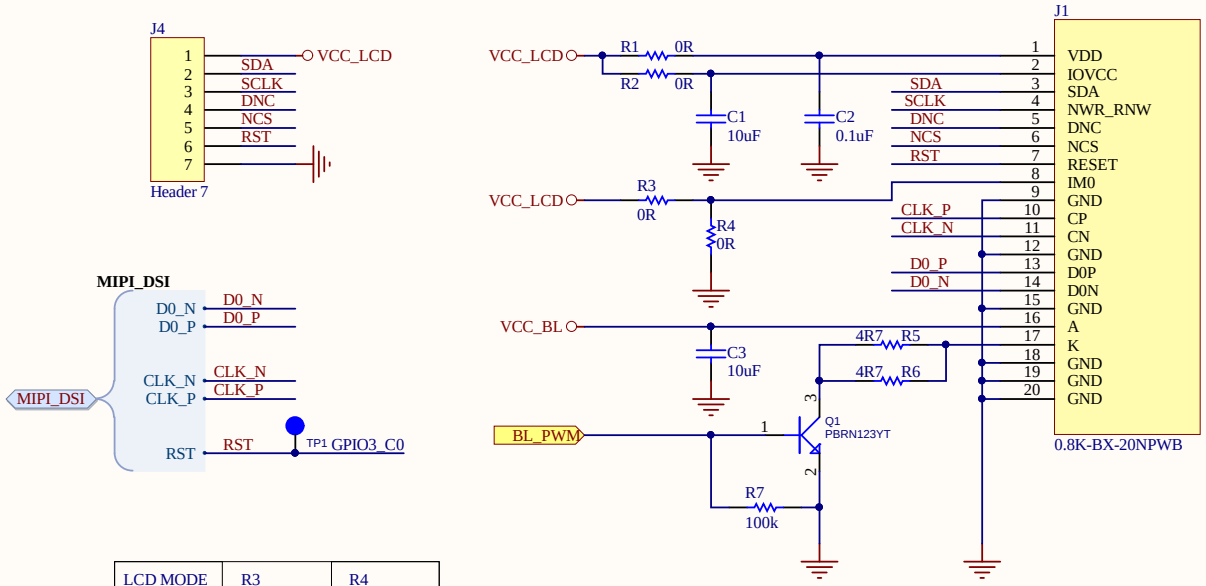



| | | | |
|---|---|---|---|
| Title | ***UART Connector*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 15. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:46 PM | Sheet 15 of 24 | | |
| File: CON_UART.SchDoc | | Designed by: ruppi kim@hardkernel.com | |




| | | | |
|---|---|---|---|
| Title | ***MIPI CSI Connector*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 16. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:46 PM | Sheet 16 of 24 | | |
| File: CON_MIPI_CAMERA.SchDoc | | Designed by: ruppi kim@hardkernel.com | |



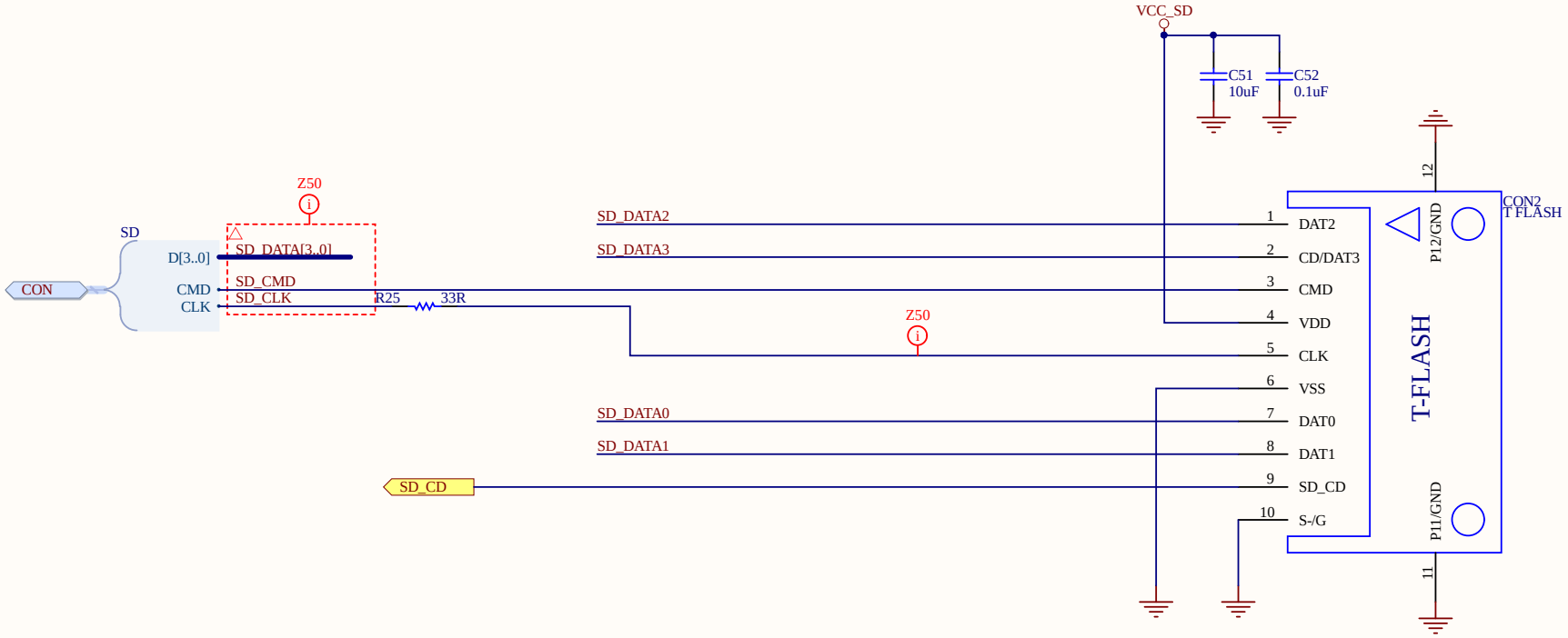

| | LCD MODE | R3 | R4 |
|---|---|---|---|
| Default | MIPI DSI | Remove | Mount |
| | 4WIRE SPI | Mount | Remove |

| | | | |
|---|---|---|---|
| Title | CON LCD KD35T133-A2 | | Hardkernel Co.,LTD.<br>475-1 Mananro<br>Manangu, Anyang City<br>Gyeonggido<br>South Korea |
| Size: A4 | Number: 17. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:47 PM | Sheet 17 of 24 | | |
| File: CON_LCD_KD35T133-A2_MIPI.SchDoc | Designed by: *@hardkernel.com | | |



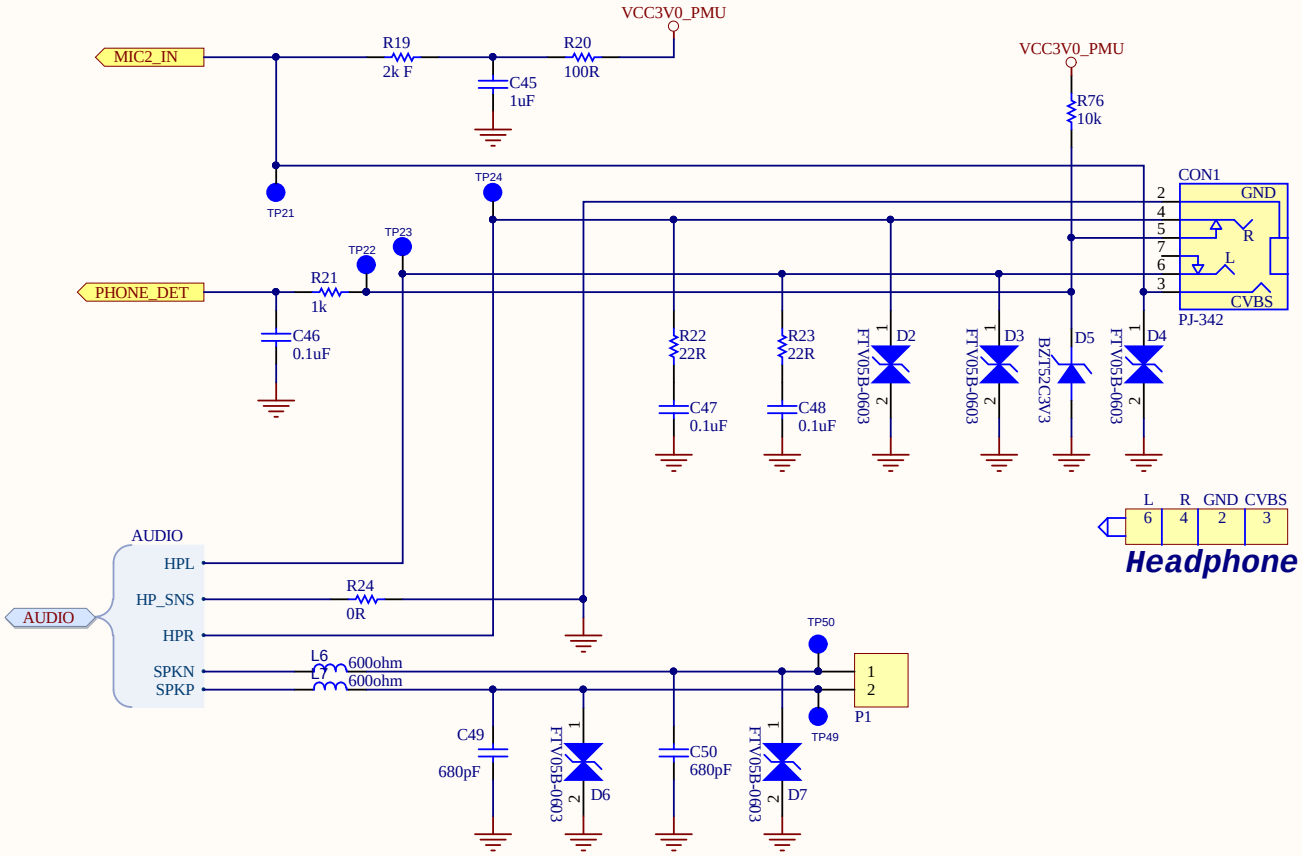

| | | | |
|---|---|---|---|
| Title | *USB 2.0 Type A Connector* | | |
| Size: A4 | Number: 18. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 18 of 24 | |
| File: CON_USB_RA.SchDoc | | | |

*Hardkernel Co.,LTD.*
*1-605 Shinan Metrokhan 1115*
*Dalandong, Dongangu, Anyang*
*Gyeonggido*
*South Korea*
Designed by: ruppi.kim@hardkernel.com




| | | | |
|---|---|---|---|
| Title | ***Micro SD Socket*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 19. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 19 of 24 | |
| File: CON_MICROSD_HD08-AL00335.SchDoc | | Designed by: ruppi kim@hardkernel.com | |

ODROID

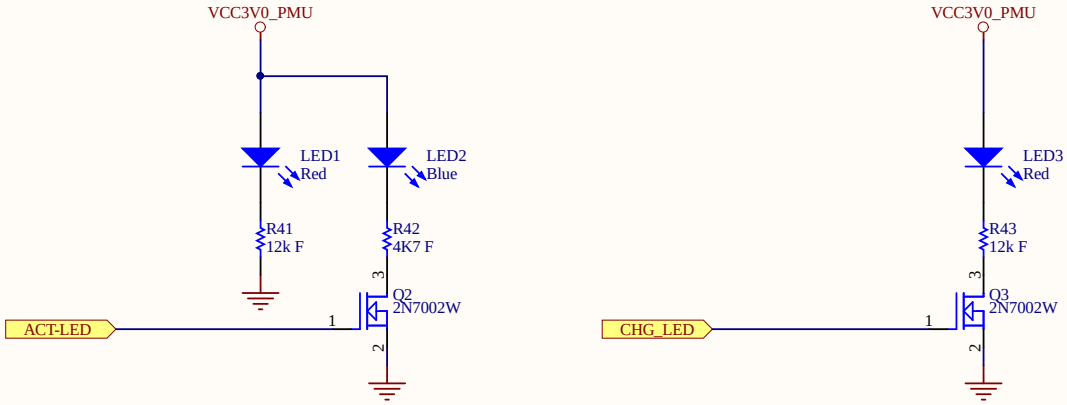



| | | | |
|---|---|---|---|
| Title | **Audio Jack** | | |
| Size: A4 | Number: 20. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 20 of 24 | |
| File: CON_AUDIO.SchDoc | | | |

*Hardkernel Co.,LTD.*
*475-1 Mananro*
*Manangu, Anyang City*
*Gyeonggido*
*South Korea*

Designed by: *@hardkernel.com

ODROID

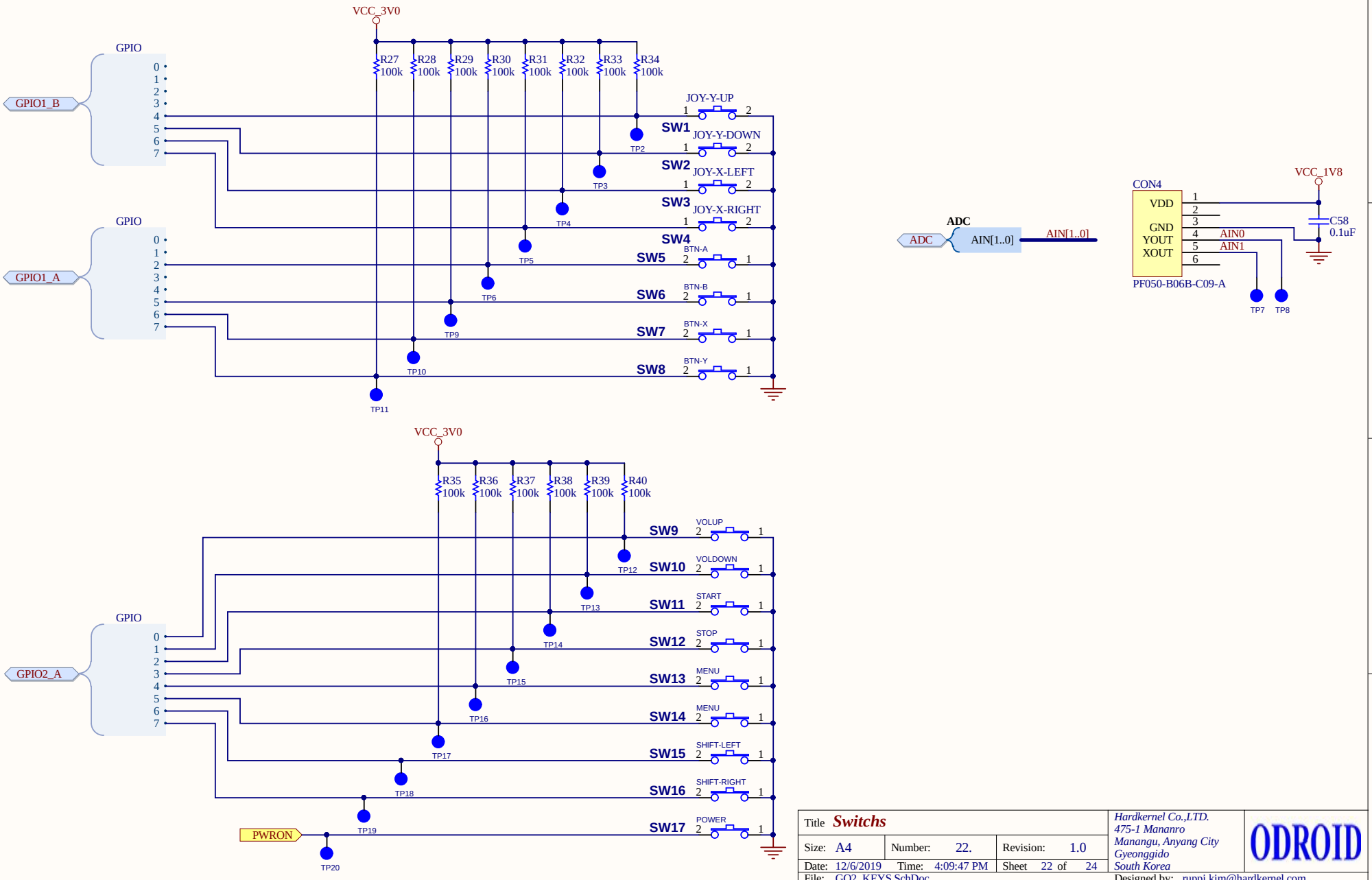



| | | | |
|---|---|---|---|
| Title | ***Indicator LEDs*** | | |
| Size: A4 | Number: 21. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 21 of 24 | |
| File: GO2_LEDS.SchDoc | | | |

*Hardkernel Co.,LTD.*
*475-1 Mananro*
*Manangu, Anyang City*
*Gyeonggido*
*South Korea*

Designed by: ruppi kim@hardkernel.com

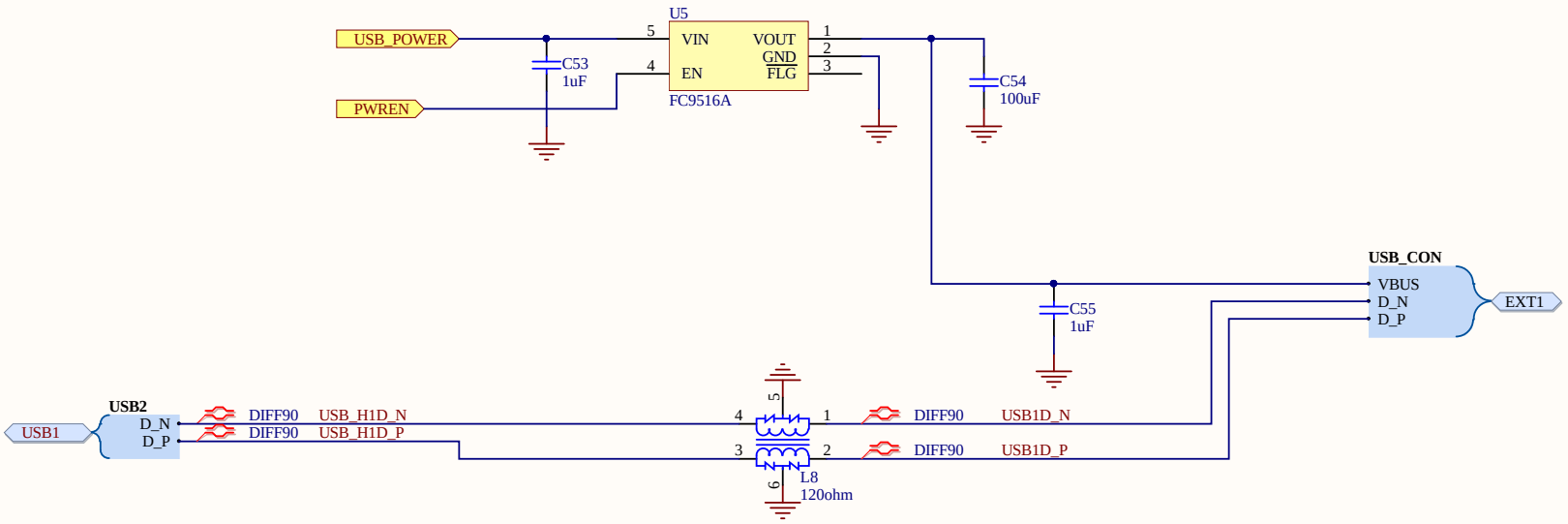
ODROID




| | | | |
|---|---|---|---|
| Title | **Switchs** | | |
| Size: A4 | Number: 22. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 22 of 24 | |
| File: GO2_KEYS.SchDoc | | Designed by: ruppi kim@hardkernel.com | |

*Hardkernel Co.,LTD.*
*475-1 Mananro*
*Manangu, Anyang City*
*Gyeonggido*
*South Korea*

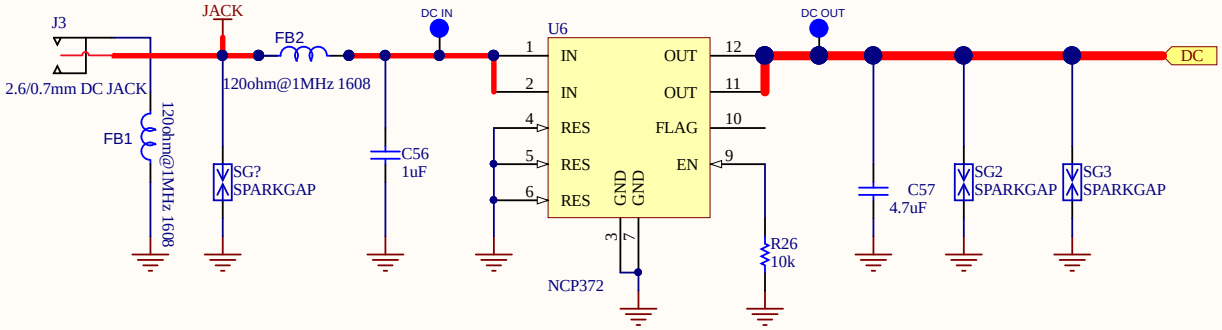
ODROID




| | | | |
|---|---|---|---|
| Title | *USB HOST PWR SW* | | |
| Size: A4 | Number: 23. | Revision: 1.0 | |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 23 of 24 | |
| File: USB_HOST_PWR_SW.SchDoc | | Designed by: ruppi.kim@hardkernel.com | |

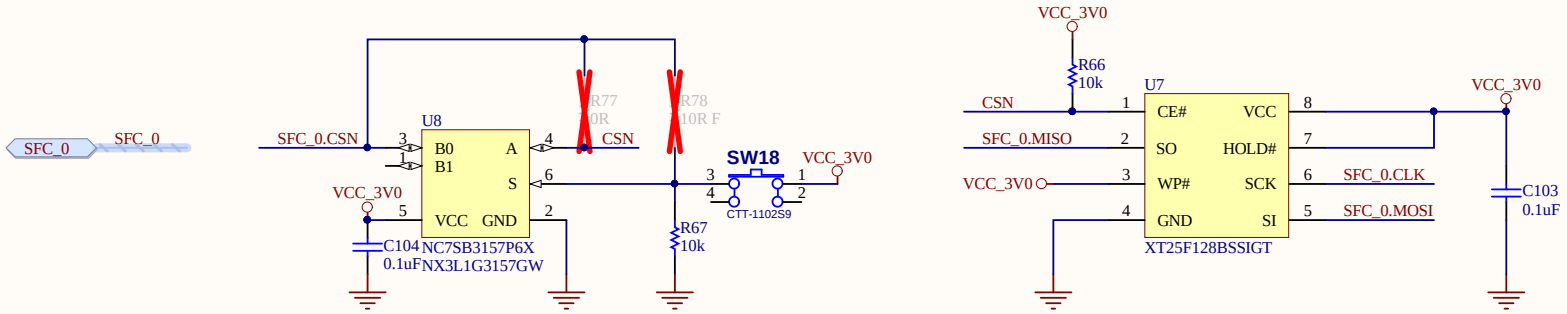
*Hardkernel Co.,LTD.*
*1-605 Shinan Metrokhan 1115*
*Dalandong, Dongangu, Anyang*
*Gyeonggido*
*South Korea*

ODROID




| | | | |
|---|---|---|---|
| Title | *CON DCJACK* | | *Hardkernel Co.,LTD.* |
| Size: A4 | Number: 24. | Revision: 1.0 | *475-1 Mananro* |
| Date: 12/6/2019 | Time: 4:09:47 PM | Sheet 24 of 24 | *Manangu, Anyang City* |
| File: CON_DCJACK.SchDoc | | Designed by: *@hardkernel.com | *Gyeonggido* |
| | | | *South Korea* |

ODROID




| | | | |
|---|---|---|---|
| Title | ***SPI FLASH MEMORY*** | | *Hardkernel Co.,LTD.*<br>*475-1 Mananro*<br>*Manangu, Anyang City*<br>*Gyeonggido*<br>*South Korea* |
| Size: A4 | Number: 25. | Revision: 1.0 | |
| Date: 12/6/2019 Time: 4:09:47 PM | Sheet 25 of 24 | | |
| File: SPI_FLASH.SchDoc | | Designed by: ruppi.kim@hardkernel.com | |